Week 3, Lecture 3
Coherence

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Notes 4

Review

What we talked about last lecture…

The squeezing operator is defined as:

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

Squeezed vacuum state:

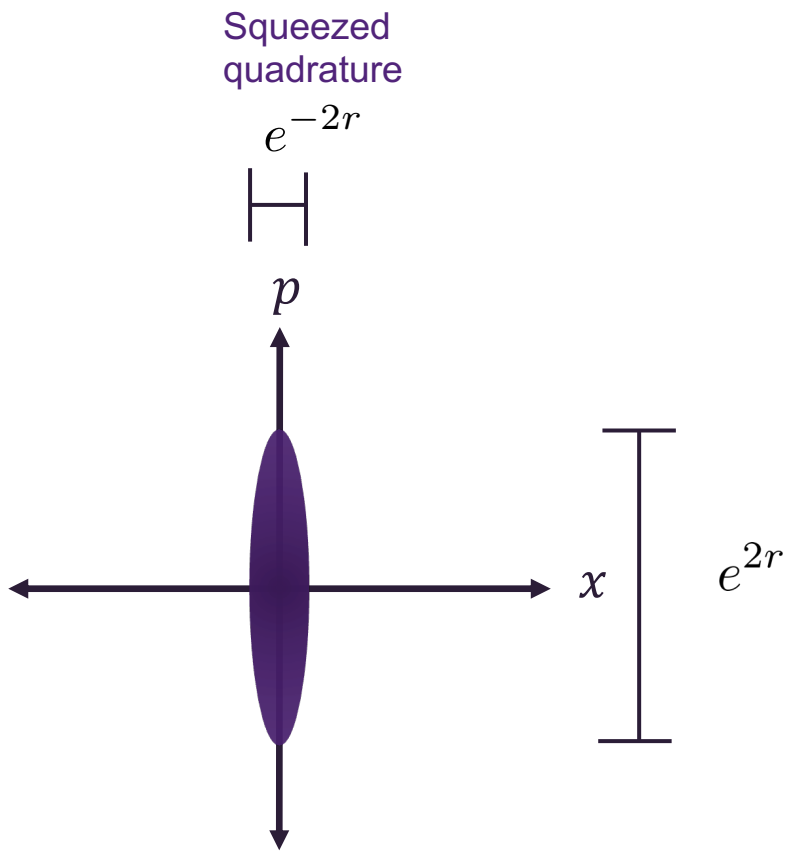
$$|\zeta\rangle = \hat{S}(\zeta)\,|0\rangle$$

General squeezed state:

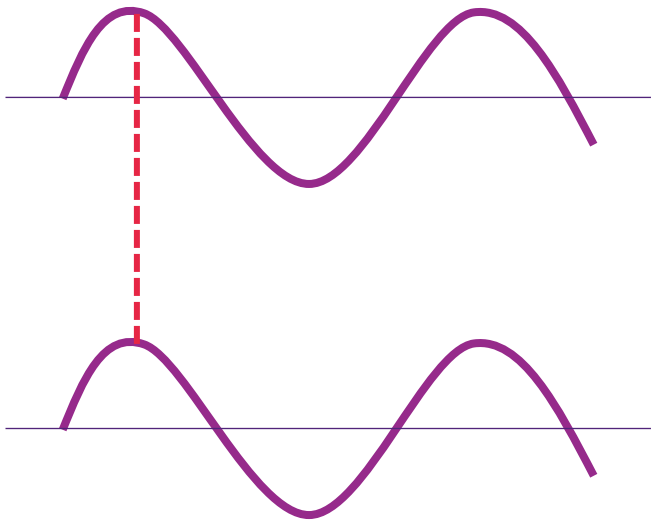
$$|\alpha, \zeta\rangle = \hat{D}(\alpha)\hat{S}(\zeta)\,|0\rangle$$

Squeezed states are minimum uncertainty states

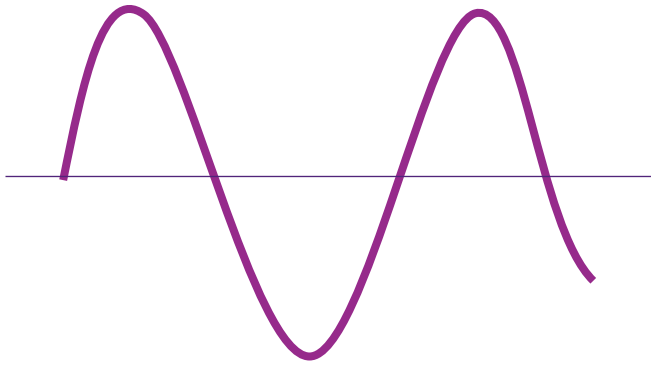
$$\langle \Delta\hat{x}^2 \rangle \langle \Delta\hat{p}^2 \rangle = e^{-2r} e^{2r} = 1$$

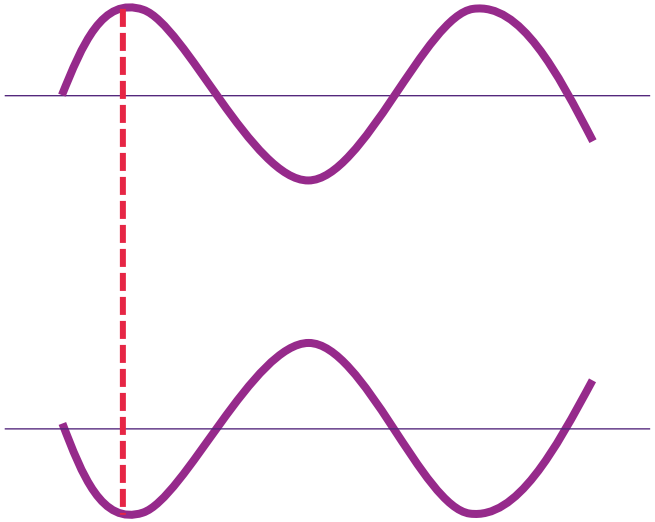


Wave interference



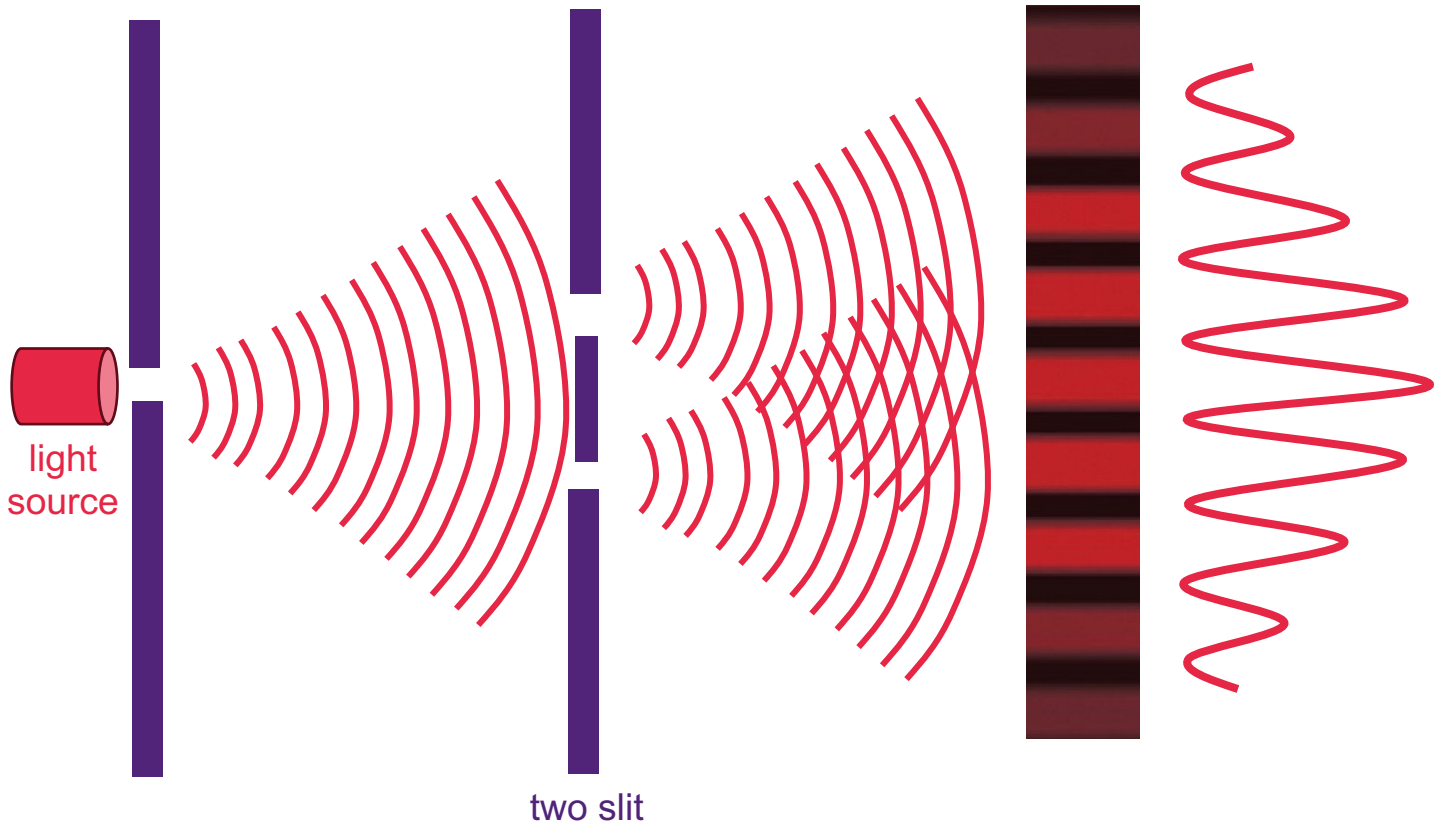
constructive interference





destructive interference

Double slit experiment



Visibility of the double slit interference requires that both slits be illuminated by a coherent wave

Coherence

Classical

Coherence describes the statistical similarity of a field at two points in space or time.

Just like the moments of number statistics, there are different orders of coherence that give different information.

First order correlation function:

$$G^{(1)}(x_1, x_2) \equiv \left\langle E^-(\mathbf{r}_1, t_1) E^+(\mathbf{r}_2, t_2) \right\rangle ,$$

First order coherence tells us about the visibility of interference fringes. Max visibility occurs for the condition of perfect first order coherence:

$$G^{(1)}(x_1, x_2) = \mathcal{E}^*(x_1)\mathcal{E}(x_2),$$

The mth order coherence function:

$$G^m(x_1 \dots x_m, y_m, \dots y_1) \equiv \left\langle E^-(x_1) \dots E^-(x_m) E^+(y_m) \dots E^+(x_1) \right\rangle .$$

Light has mth order coherence if this function factorises into the product of amplitudes:

$$G^m(x_1 \dots x_m, y_m, \dots y_1) \equiv \mathcal{E}^*(x_1) \dots \mathcal{E}^*(x_m)\mathcal{E}(y_m) \dots \mathcal{E}(x_1).$$

See Lecture notes Chapter 4

Second order correlation function

Classical

A useful measure for quantifying fluctuations in intensity of light is the **second order correlation function**

$$\begin{aligned}\langle I(t)I(t+\tau)\rangle &= \left\langle E^{-}(t)E^{-}(t+\tau)E^{+}(t+\tau)E^{+}(t)\right\rangle \\ &= G^{(2)}(t,t+\tau,t+\tau,t) \equiv G^{(2)}(\tau),\end{aligned}$$

Light has second order coherence for the condition:

$$G^{(2)}(\tau) = \mathcal{E}^{*}(0)\mathcal{E}^{*}(\tau)\mathcal{E}(\tau)\mathcal{E}(0),$$

We can use the normalised second order coherence function:

$$g^{(2)}(\tau) \equiv \frac{G^{(2)}(\tau)}{\left[G^{(1)}(0)\right]^{2}} = \frac{\langle I(t)I(t+\tau)\rangle}{\langle I\rangle^{2}}$$

Mathematically, we can relate $g^{(2)}(0)$ to the variance of the intensity fluctuations:

$$\begin{aligned}g^{(2)}(0) &= \frac{\langle I^{2}\rangle}{\langle I\rangle^{2}} = 1 + \frac{\langle I^{2}\rangle - \langle I\rangle^{2}}{\langle I\rangle^{2}} \\ &= 1 + \frac{\mathrm{var}(I)}{\langle I\rangle^{2}}.\end{aligned}$$

classical result
$g^{(2)}(0)$ cannot be less than 1

Second order correlation function

Quantum

To obtain the quantum result, consider $g^{(2)}(0)$ for a single-mode state:

$$\begin{aligned} g^{(2)}(0) &= \frac{\langle a^\dagger a^\dagger a a\rangle}{\langle a^\dagger a\rangle\langle a^\dagger a\rangle} \\ &= \frac{\langle a^\dagger a a^\dagger a\rangle - \langle a^\dagger a\rangle}{\langle a^\dagger a\rangle\langle a^\dagger a\rangle} \\ &= 1 + \frac{\mathrm{var}(n) - \bar{n}}{\bar{n}^2} \end{aligned}$$

Thermal state

$$\mathrm{var}\,(n) = \langle \hat{n}^2\rangle - \langle \hat{n}\rangle^2 = \bar{n}^2 + \bar{n} \qquad g^{(2)} = 2$$

Coherent state

$$\mathrm{var}\,(n) = \langle \hat{n}^2\rangle - \langle \hat{n}\rangle^2 = \bar{n} \qquad g^{(2)} = 1$$

Coherent states have coherence in all orders

Number state

$$\mathrm{var}\,(n) = \langle \Delta\hat{n}^2\rangle = \langle \hat{n}^2\rangle - \langle \hat{n}\rangle^2 = 0 \qquad g^{(2)} = 1 - \frac{1}{n} < 1$$

$g^{(2)}(0) < 1$ has no classical explanation

Photon arrival statistics

Photon bunching

$$g^{(2)}(0) > 1$$

Photons are more likely to arrive close together than for independent arrivals.
Example: thermal light $g^{(2)}(0) = 2$

Independent arrivals

$$g^{(2)}(0) = 1$$

Detecting one photon does not affect the likelihood of detecting another.
Example: coherent light.

Photon antibunching

$$g^{(2)}(0) < 1$$

Photons are less likely to arrive close together than for independent arrivals.
Example: an ideal single-photon state $g^{(2)}(0) = 0$

Antibunching is a signature of nonclassical light.

Fano factor

The ratio of the variance to the mean is called the Fano factor:

$$F = \frac{\mathrm{var}(n)}{\langle n \rangle}$$

Thermal state

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \bar{n}^2 + \bar{n}$$

$F > 1$ Super-poissonian

Coherent state

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \bar{n}$$

$F = 1$ Poissonian

Number state

$$\mathrm{var}\,(n) = \langle \Delta \hat{n}^2 \rangle = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = 0$$

$F < 1$ Sub-poissonian

Summary of classification by photon statistics

	$\langle \hat{n}^2 \rangle$	**Photon number statistics**	$g^{(2)}(0)$	**Fano factor**
Thermal	$\bar{n}(1+2\bar{n})$	Super poissonian	2	>1
Coherent	$\bar{n}(1+\bar{n})$	Poissonian	1	1
$\|n\rangle$	n^2	Sub-poissonian	$1-\frac{1}{n}$	<1

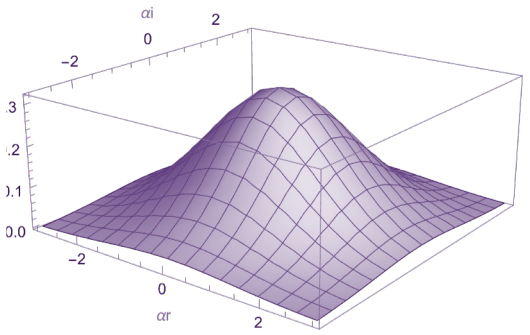
Gaussian states

Note: material on the following slides, up to and including Slide 22 will not be examined.

Gaussian states are defined as states with a Gaussian Wigner function

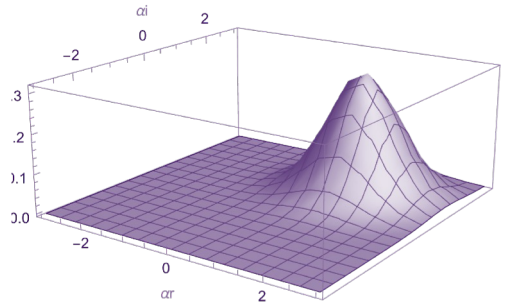
Gaussian operations preserve this form

Thermal states



$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1} \right)^n |n\rangle \langle n|$$

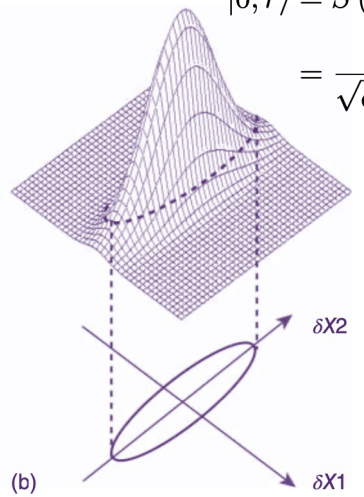
Coherent states



$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Squeezed states

$$|0, r\rangle = S(r)|0\rangle$$
$$= \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} \tanh r^n |2n\rangle$$



Gaussian states

Gaussian states are completely characterised by their mean vector d and covariance matrix Σ (first and second moments)

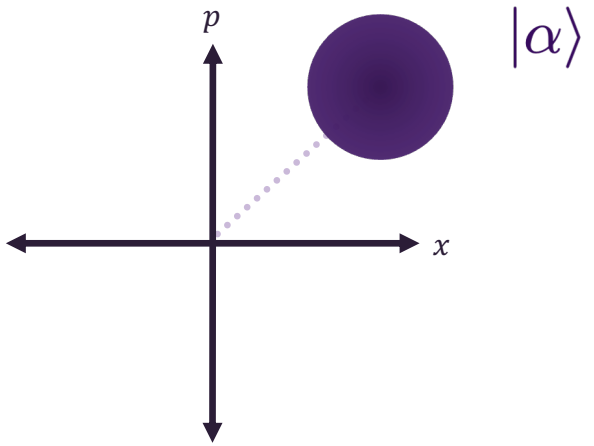
Covariance matrix is defined as

$$\Sigma_{ij} := \frac{1}{2}\langle\{\Delta\hat{x}_i, \Delta\hat{x}_j\}\rangle$$
$$\Delta\hat{x}_i := \hat{x}_i - \langle\hat{x}_i\rangle$$

diagonal elements of the covariance matrix provide the variances of the quadrature operators

Mean vector

$$\overline{\mathrm{d}} = \langle\hat{\boldsymbol{d}}\rangle = \mathrm{Tr}\left[\hat{\rho}\hat{\boldsymbol{d}}\right]$$



Coherent state

Covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

var (x)

var (p)

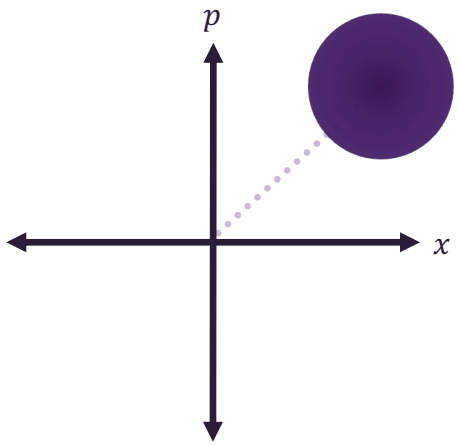
Mean vector

$$\mathrm{d} = (2\mathrm{Re}(\alpha), 2\mathrm{Im}(\alpha))$$

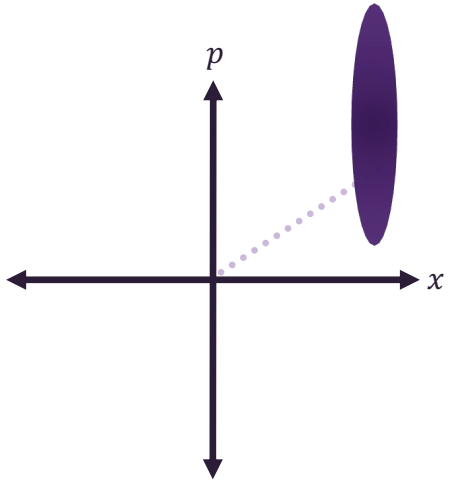
Gaussian states

$$\Sigma_{ij} := \frac{1}{2}\langle\{\Delta\hat{x}_i, \Delta\hat{x}_j\}\rangle$$
$$\Delta\hat{x}_i := \hat{x}_i - \langle\hat{x}_i\rangle$$

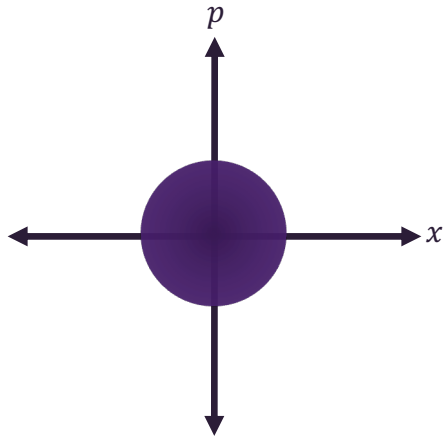
Gaussian states are completely characterised by their mean vector d and covariance matrix Σ (first and second moments)



Coherent state



Squeezed state



Vacuum state

Covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

var (x)

var (p)

$$\Sigma = \begin{bmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Mean vector

$$\mathrm{d} = (2\mathrm{Re}(\alpha), 2\mathrm{Im}(\alpha))$$

$$\mathrm{d} = (x, p)$$

$$\mathrm{d} = (0, 0)$$

Transformation of Gaussian states

- Gaussian operations are defined as operations that map Gaussian states to Gaussian states.
- For Gaussian inputs, the outputs are also Gaussian This means the outputs are completely characterised by their mean vector d and covariance matrix Σ
- A beam-splitter is an example of a Gaussian operation

Gaussian unitaries transform the mean vector as

$$d_{\mathrm{out}} = S d_{\mathrm{in}} + y \qquad y \in \mathbb{R}^{2n}$$

And the covariance matrix as

$$\Sigma_{\mathrm{out}} = S \Sigma_{\mathrm{in}} S^T$$

Where S is a symplectic matrix

S is symplectic if

$$\mathbf{S}\boldsymbol{\Omega}\mathbf{S}^T = \boldsymbol{\Omega}\,.$$

$$\boldsymbol{\Omega} := \bigoplus_{k=1}^{N} \boldsymbol{\omega} = \begin{pmatrix} \boldsymbol{\omega} & & \\ & \ddots & \\ & & \boldsymbol{\omega} \end{pmatrix} \quad \boldsymbol{\omega} := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

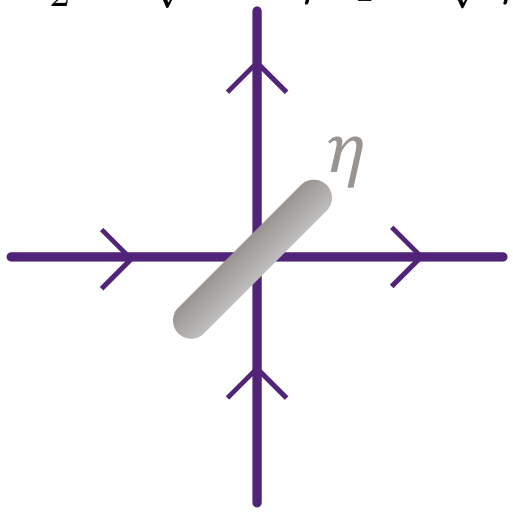
For more information, see:
C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro, and S. Lloyd, *Gaussian Quantum Information*, Rev. Mod. Phys. **84**, 621 (2012).

Gaussian operations

Examples of Gaussian operations:

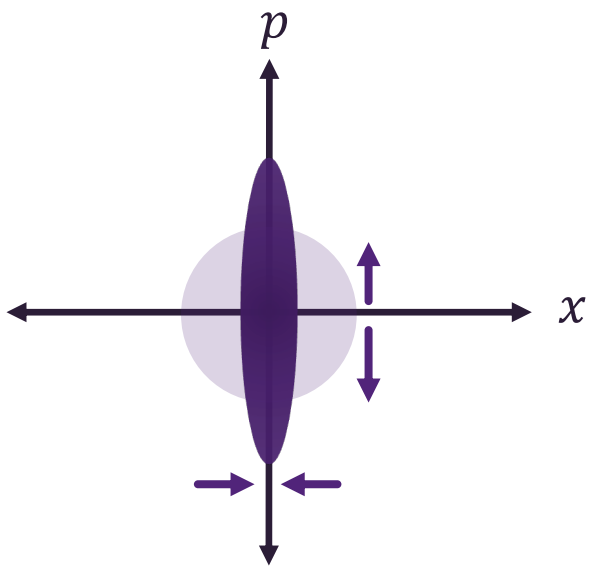
Beam splitters

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$
$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$



Single mode squeezing

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$



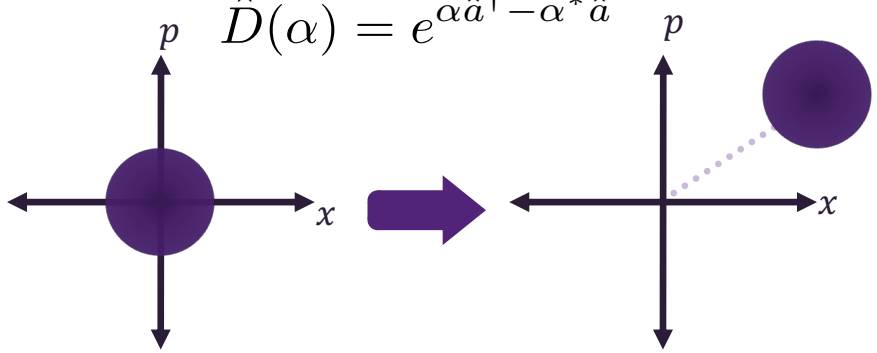
Phase shift

$$\hat{U}(\theta) = \exp(-i\theta\hat{n})$$



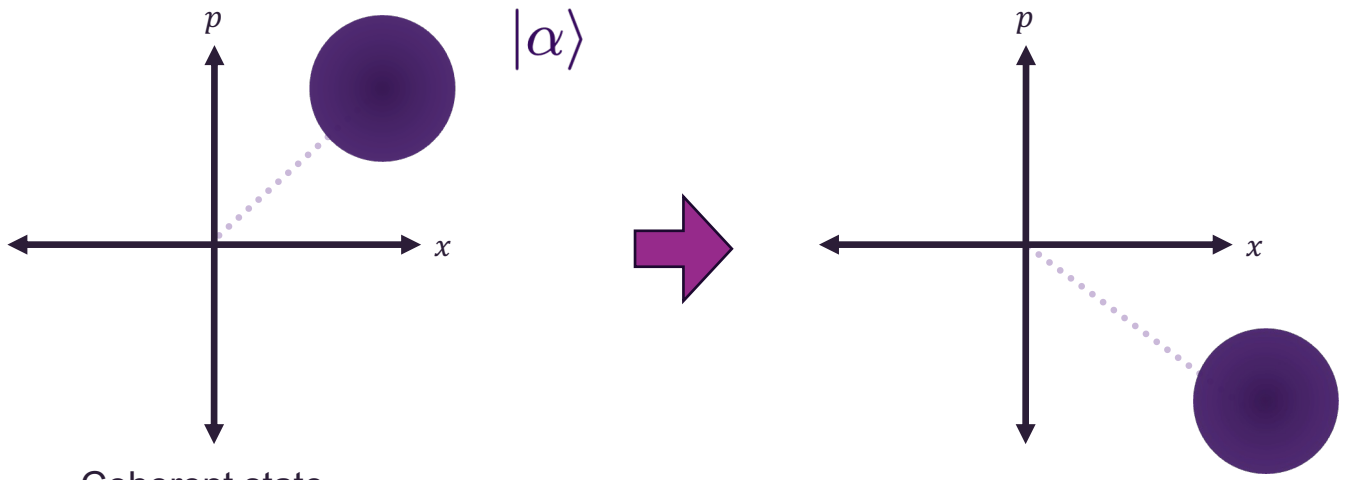
Displacement

$$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$$



Displacement

Example: Displacement operator on a coherent state $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$



Coherent state

Covariance matrix $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ var (x) var (p)

Mean vector $\mathrm{d} = (2\mathrm{Re}(\alpha), 2\mathrm{Im}(\alpha))$

Displacement operator

$\hat{D}(\beta) = e^{\beta \hat{a}^\dagger - \beta^* \hat{a}}$

Using transformation of mean vector and covariance matrix

$$d_{\text{out}} = S d_{\text{in}} + y$$

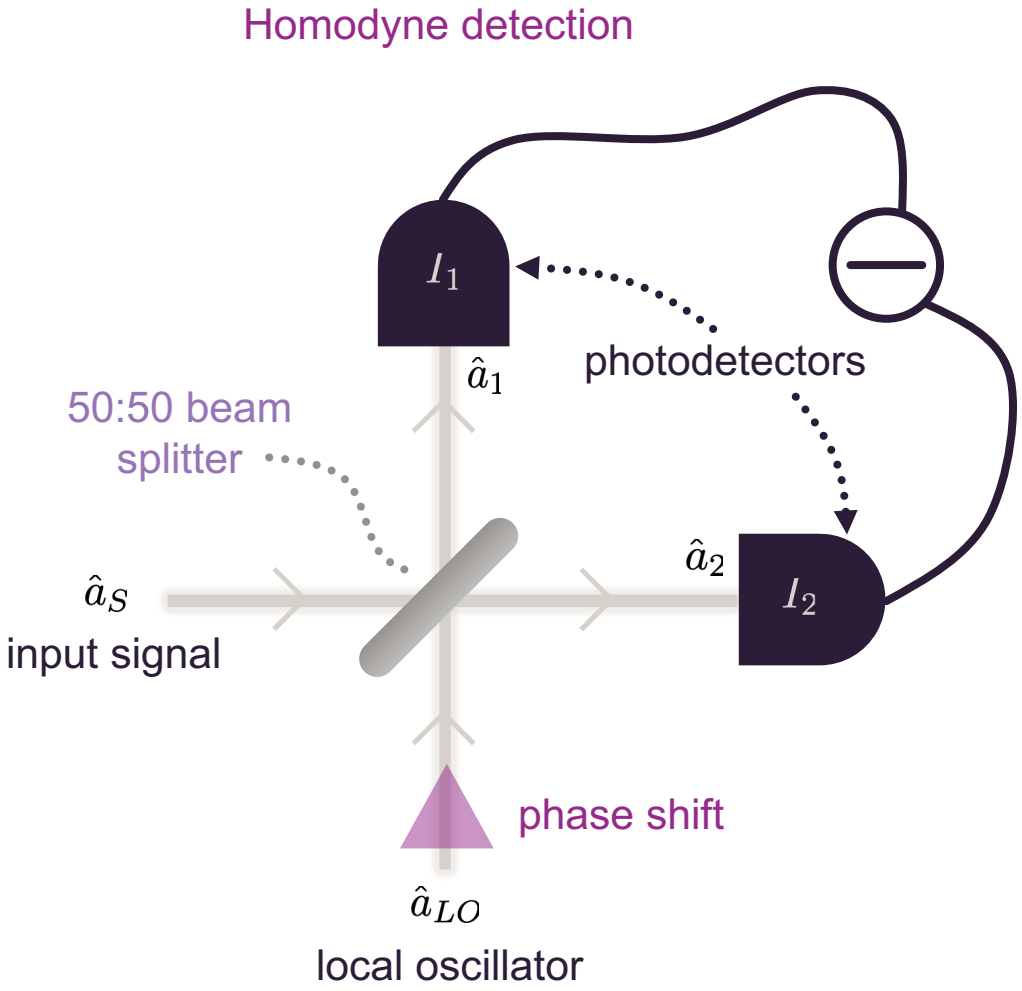
$$\Sigma_{\text{out}} = S \Sigma_{\text{in}} S^T$$

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad y = (x, p)$$

$$d_{\text{out}} = d_{\text{in}} + y$$
$$= (2\mathrm{Re}(\alpha) + x, 2\mathrm{Im}(\alpha) + y)$$

$$\Sigma_{\text{out}} = \Sigma_{\text{in}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Gaussian measurement



Beam splitter in symplectic form

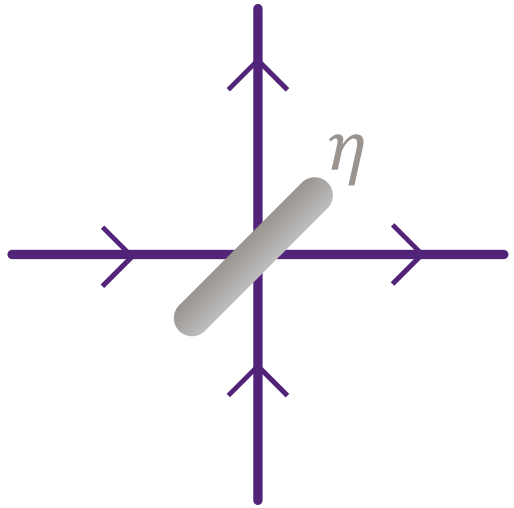
- The matrix S_{BS} is a 4×4 matrix as it operates on two Gaussian modes.
- Here, η is the transmission of the beam splitter.

$$d_{\text{out}} = S d_{\text{in}} + y$$

$$\Sigma_{\text{out}} = S \Sigma_{\text{in}} S^T$$

$$S_{\text{BS}} = \underbrace{\begin{bmatrix} \sqrt{\eta} I_2 & \sqrt{1-\eta} I_2 \\ -\sqrt{1-\eta} I_2 & \sqrt{\eta} I_2 \end{bmatrix}}_{4\times4 \text{ matrix}}$$

$$y = (0, 0, 0, 0)$$



where η is the transmission of the beamsplitter,

I_2 is the 2×2 identity matrix

Squeezed states

- Using the symplectic form of the beam splitter operation S_{BS}, we can easily find the state of the two output modes of a beam splitter

Example: what is the effect of optical loss on a squeezed vacuum state?

- Model this via a beam splitter mixing a squeezed vacuum state with a vacuum state

Squeezed vacuum state

$$\Sigma_{\mathrm{sq}} = \begin{bmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{bmatrix} \qquad \Sigma_{\mathrm{vac}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathrm{d}_{\mathrm{sq}} = (0,0) \qquad \mathrm{d}_{\mathrm{vac}} = (0,0)$$

CM of two modes before beam splitter

$$\Sigma = \begin{bmatrix} \Sigma_{\mathrm{sq}} & 0 \\ 0 & \Sigma_{\mathrm{vac}} \end{bmatrix} = \begin{bmatrix} e^{-2r} & 0 & 0 & 0 \\ 0 & e^{2r} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$S_{\mathrm{BS}} = \begin{bmatrix} \sqrt{\eta} I_2 & \sqrt{1-\eta} I_2 \\ -\sqrt{1-\eta} I & \sqrt{\eta} I_2 \end{bmatrix}$$

$$y = (0,0,0,0)$$

Squeezed states

CM of two modes after beam splitter

$$\Sigma_{\text{out}} = S_{\text{BS}} \Sigma_{\text{in}} S_{\text{BS}}^{T}$$

$$= \begin{bmatrix} \eta e^{-2r} + (1-\eta) & 0 & e^{-2r}\left(e^{2r}-1\right)\sqrt{\eta\,(1-\eta)} & 0 \\ 0 & \eta e^{2r} + (1-\eta) & 0 & \left(1-e^{2r}\right)\sqrt{\eta\,(1-\eta)} \\ e^{-2r}\left(e^{2r}-1\right)\sqrt{\eta\,(1-\eta)} & 0 & \eta + (1-\eta)\,e^{-2r} & 0 \\ 0 & \left(1-e^{2r}\right)\sqrt{\eta\,(1-\eta)} & 0 & \eta + (1-\eta)\,e^{2r} \end{bmatrix}$$

- With the 4×4 covariance matrix of the two output modes of the beam splitter, take the partial trace over the environment mode to get the reduced state of the squeezed state after it has passed through loss
- To take partial trace of covariance matrices, simply remove rows and columns of excluded modes

$$\Sigma = \begin{bmatrix} \eta e^{-2r} + (1-\eta) & 0 \\ 0 & \eta e^{2r} + (1-\eta) \end{bmatrix}$$

Squeezed states

CM after loss

$$\Sigma = \begin{bmatrix} \eta e^{-2r} + (1-\eta) & 0 \\ 0 & \eta e^{2r} + (1-\eta) \end{bmatrix}$$

Compare this to covariance matrix before loss

$$\Sigma = \begin{bmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{bmatrix}$$

Variance of squeezed quadrature

$$e^{-2r} \rightarrow \eta e^{-2r} + (1-\eta)$$

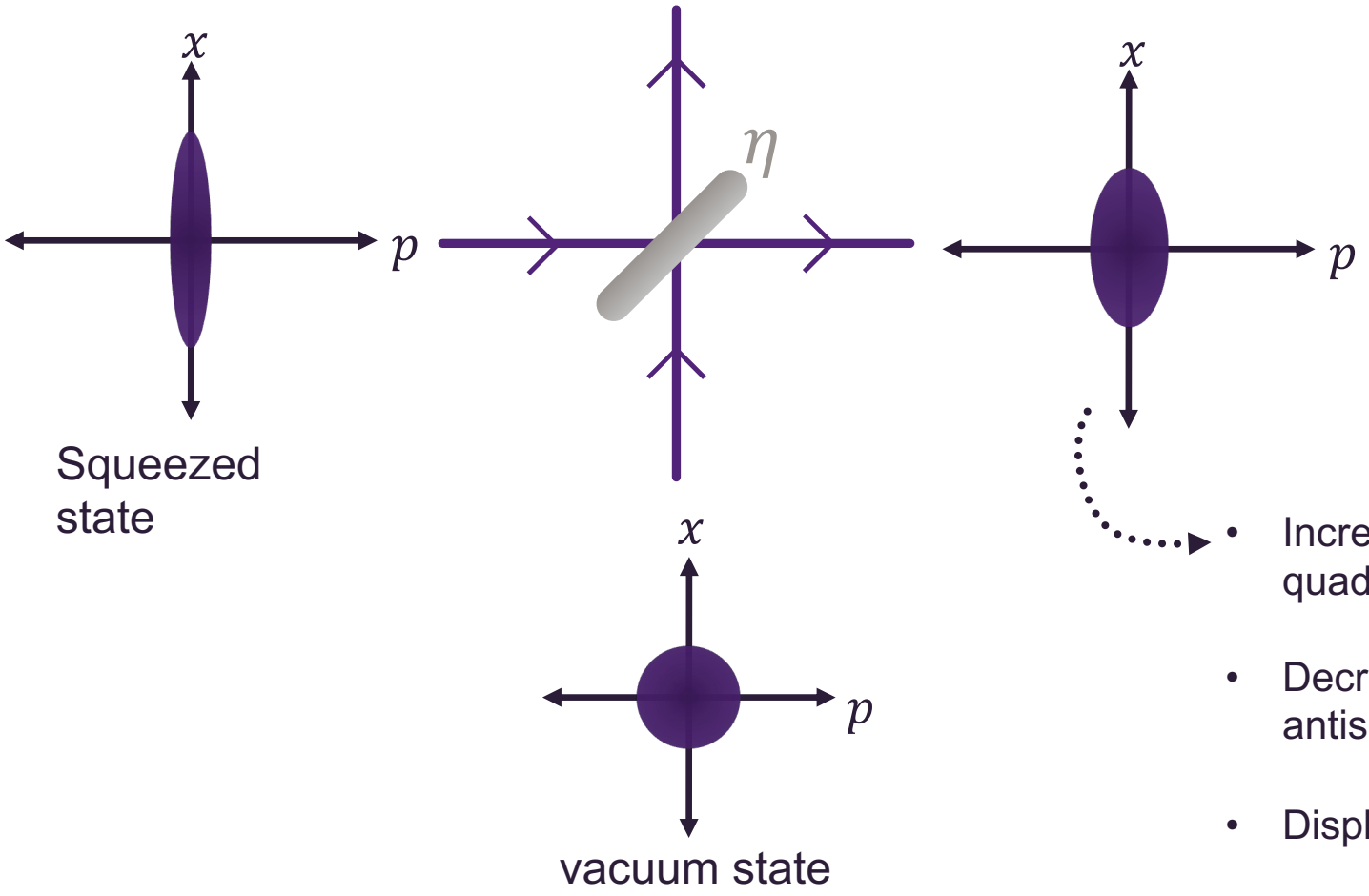
Greater than e^{-2r} for $\eta \in [0,1]$

Variance of anti-squeezed quadrature

$$e^{2r} \rightarrow \eta e^{2r} + (1-\eta)$$

Less than e^{2r} for $\eta \in [0,1]$

Squeezed states



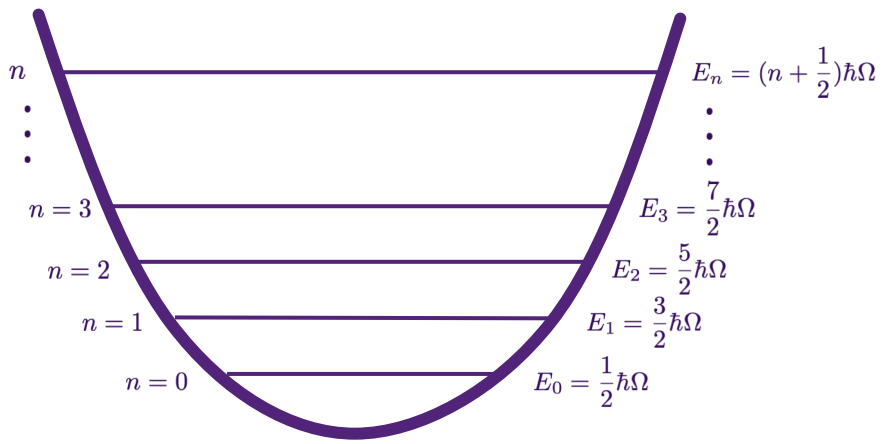
After a squeezed state passes through loss, the variance of the squeezed quadrature increases while the variance of the anti-squeezed quadrature decreases.

- Increase in variance of squeezed quadrature
- Decrease in variance of antisqueezed quadrature
- Displacement still at (0,0)

less like an ellipse more like a sphere

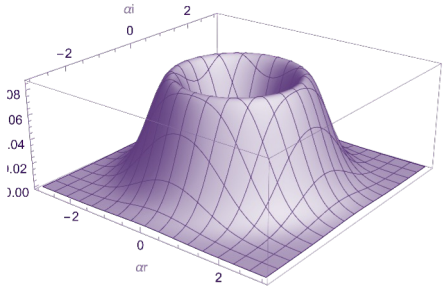
Review

Quantum states



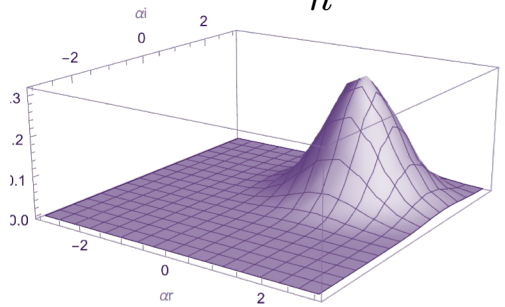
Fock photon number states

$$\{|n\rangle\} = \{|0\rangle, |1\rangle, |2\rangle, \cdots\}$$



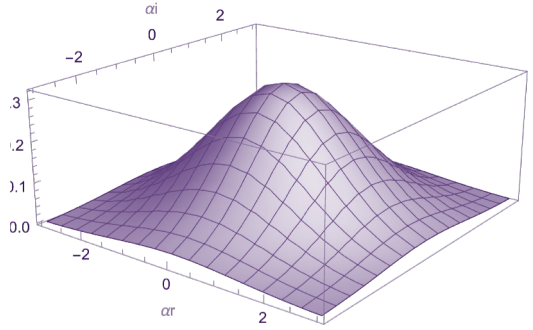
Coherent states

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$



Thermal states

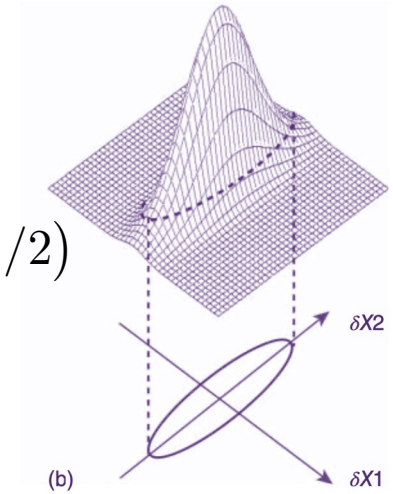
$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n |n\rangle\langle n|$$



Squeezed states

$$S(\zeta) = \exp\left(\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right)/2\right)$$

$$|r\rangle = \hat{S}(r)|0\rangle$$

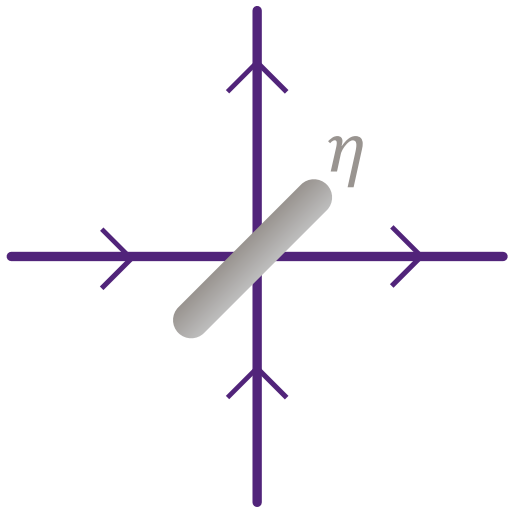


Operations

Beam splitters

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$

$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$

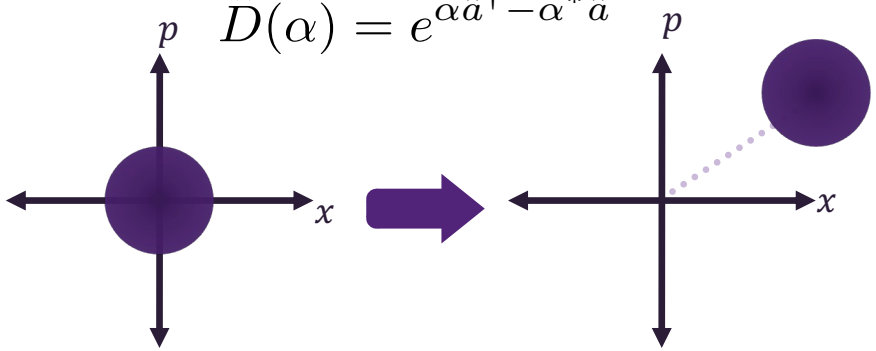


Phase shift

$$\hat{U}(\theta) = \exp(-i\theta\hat{n})$$

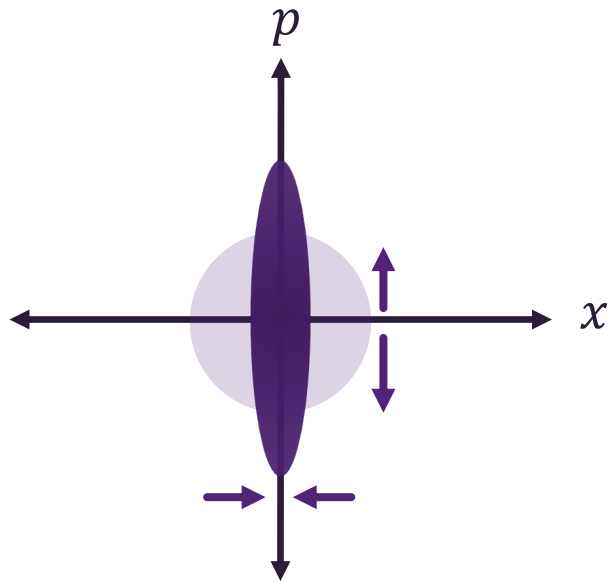
Displacement

$$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$$



Single mode squeezing

$$S(\zeta) = \exp\left(\left(\zeta^*\hat{a}^2 - \zeta\hat{a}^{\dagger 2}\right)/2\right)$$



Measurement

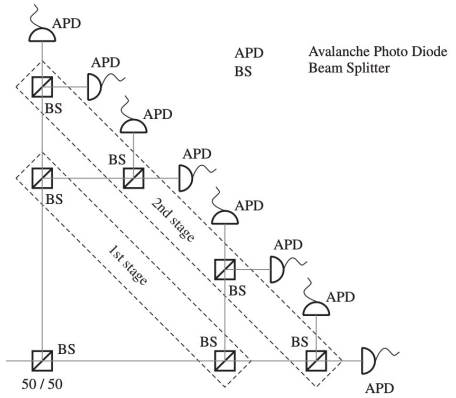
Detecting single photons

Avalanche Photodiodes

$$\Pi_0 = |0\rangle\langle 0| \qquad \Pi_1 = \mathbb{1} - |0\rangle\langle 0|$$

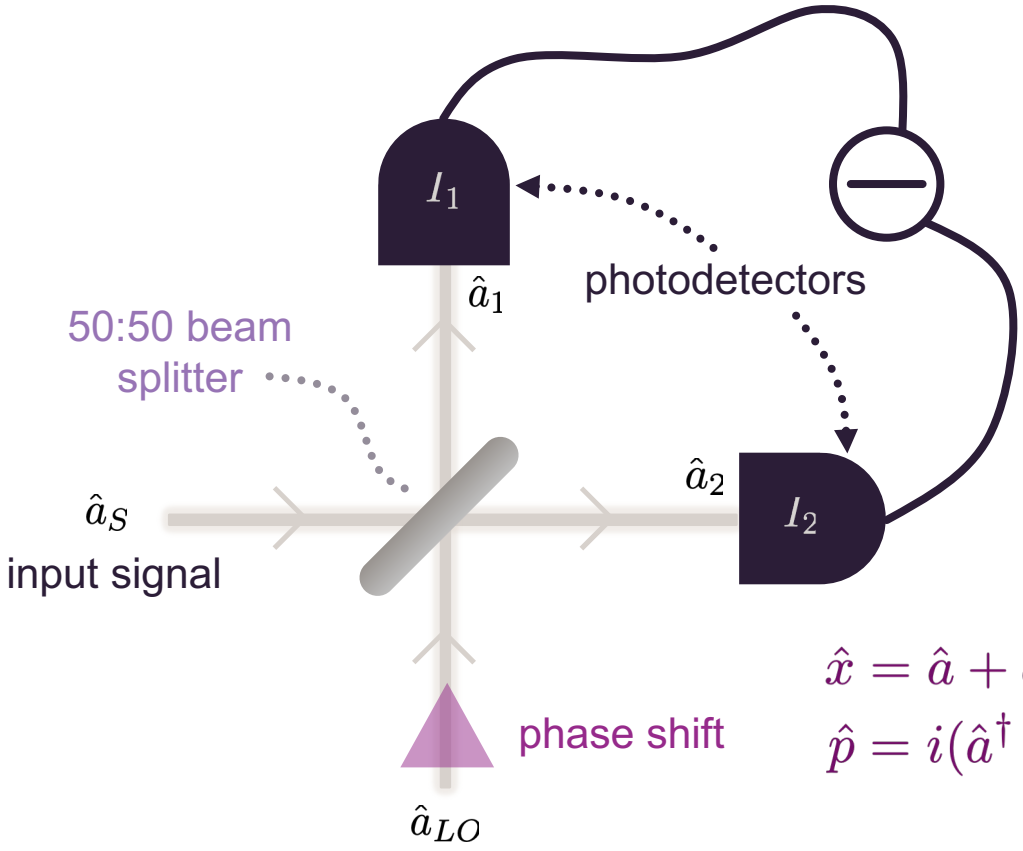


Resolving photon number



Measuring quadratures

Homodyne detection



Visualising quantum states

P function

For coherent states:

$$P(\alpha) = \delta^2(\alpha - \beta)$$

Delta function

i.e. a single point in phase space

Wigner function

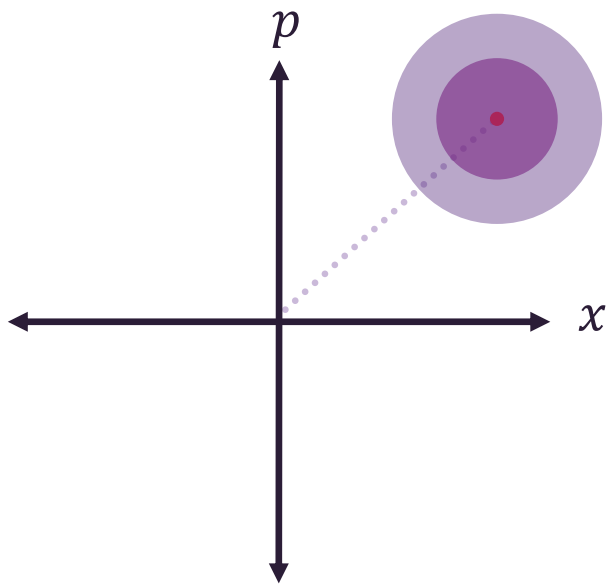
$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha-\beta|^2}$$

Gaussian

Q function

$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha-\beta|^2}$$

Broader Gaussian



Summary of classification by photon statistics

	$\langle \hat{n}^2 \rangle$	**Photon number statistics**	$g^{(2)}(0)$	**Fano factor**
Thermal	$\bar{n}(1+2\bar{n})$	Super poissonian	2	>1
Coherent	$\bar{n}(1+\bar{n})$	Poissonian	1	1
$\|n\rangle$	n^2	Sub-poissonian	$1-\frac{1}{n}$	<1